



Republic of the Philippines
Department of Education
REGION III
SCHOOLS DIVISION OFFICE OF NUEVA ECija

LEARNING ACTIVITY SHEET
SEPCIAL PROGRAM IN ICT 10
COMPUTER SYSTEM SERVICING
First Quarter, Week 1

Name of Learner: _____

Date: _____

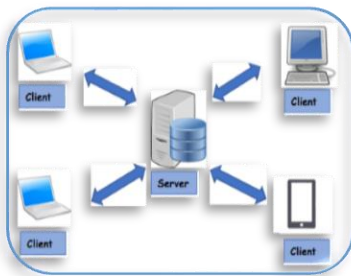
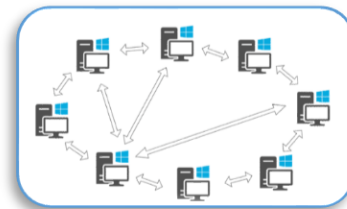
Grade Level /Section: _____

Network Operating System (NOS)

BACKGROUND INFORMATION FOR LEARNERS

Network operating system abbreviated as NOS is a type of operating system designed primarily to perform the function of supporting workstations, personal computers and older terminals connected on local area network also known as LAN. The software that makes NOS works enable the different devices attached to it to communicate with each other. There are two types of network operating system, peer-to-peer and client server.

1. **Peer-to-peer NOS** allow the users to share networks in a common network location. All devices are equal in terms of its functions and uses. It is best fitted for small to medium LAN and is cheaper compared to the other type of NOS.



2. **Client-server NOS** gives the users access to resources that are saved through a server. The functions and applications are unified in one file server and can be used to perform single client actions notwithstanding the physical location. It tends to be more expensive than the other type of NOS and needed a huge amount to technical maintenance. It's advantage over peer-to-peer NOS is that it can be centrally controlled and is easier to incorporate with other technology.



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COMPARISON BETWEEN CLIENT-SERVER AND PEER-TO-PEER NOS

BASIS FOR COMPARISON	CLIENT-SERVER	PEER-TO-PEER
Basic	There is a particular server and clients that are connected to the server.	Clients and server are not identified; each computer can act as client and server.
Service	The client ask/command for service and server respond with the command.	Each computer can command for services and can also respond to the command of service.
Focus	The information is being shared.	The information is connected with each other
Data	With a centralized server where the data are stored.	Each peer has its own data.
Server	Congestion of process may happen when several clients request for the services simultaneously	Congestion of process will not happen because the services are provided by several servers distributed in the peer-to-peer system
Expense	expensive	Cheaper
Stability	stable and scalable	Stability decreases as the number of peer increases.

LEARNING COMPETENCY

1. Set-up user access.

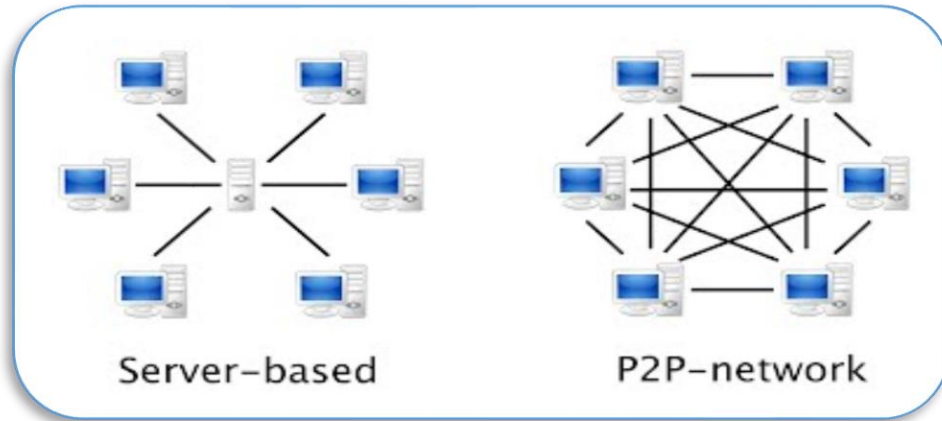


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ACTIVITIES

ACTIVITY 1

Direction: Analyze the pictures below, then answer the questions that follows:



Guide Questions:

1. What is network operating system?

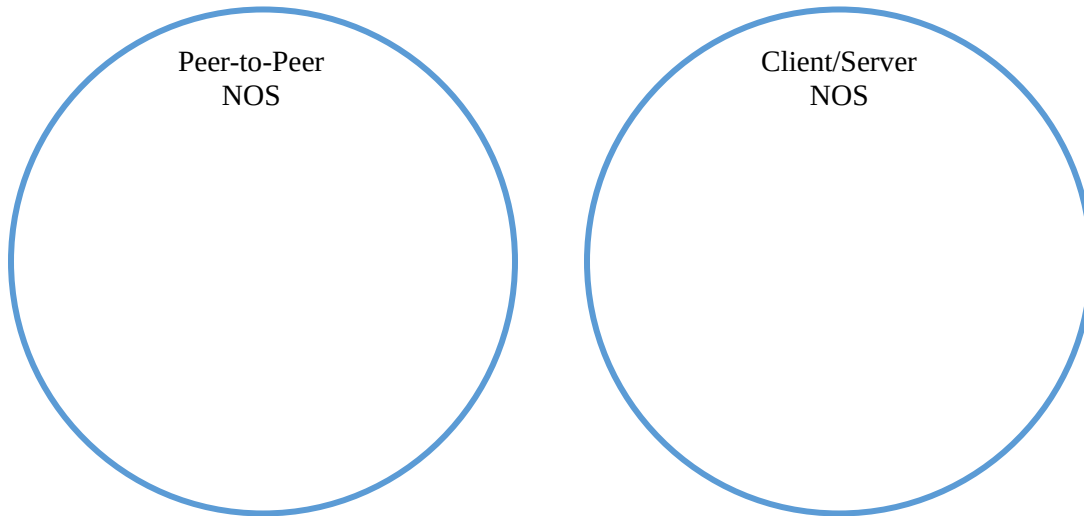
2. Based on the picture above, describe the peer-to-peer NOS.

3. Give the salient characteristics of client/server based NOS.



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4. Give the differences between the two type of NOS.



RUBRICS FOR SCORING

- 5- the answer is strongly relevant to the question
- 4- the answer is relevant to the question
- 3- the answer is nearly relevant to the question
- 2- the answer is nearly irrelevant to the question
- 1 –the answer is irrelevant to the question

ACTIVITY 2

Direction: Analyze the following statements then write **P2P** if it pertains to peer-to-peer NOS and **CS** if it pertains to client-server NOS.

- _____1. Stability decreases as the number of peer increases.
- _____2. Congestion of process may happen when several clients request for the services simultaneously
- _____3. The client ask/command for service and server respond with the command.
- _____4. Each computer can command for services and can also respond to the command of service.
- _____5. There is a particular server and clients that are connected to the server.



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- _____6. Congestion of process will not happen because the services are provided by several servers distributed in the peer-to-peer system
- _____7. Each computer can command for services and can also respond to the command of service.
- _____8. stable and scalable
- _____9. cheaper
- _____10. The information is being shared.

REFLECTION:

What is the importance of network operating system?

REFERENCES FOR LEARNERS

<https://searchnetworking.techtarget.com/definition/network-operating-system>

https://www.google.com/search?q=peer+to+peer+network&sxsrf=ALeKk02znDm6EJr2BoaC4MW_AyZ_uXlCuA:1600692555034&tbm=isch&source=iu&ictx=1&fir=LuL2X71fAQh16M%252Cs-1BXys4xVy91M%252C_&vet=1&usg=AI4_-kRtsX72IJwd7dyoz6X1qy0gFiQxPw&sa=X&ved=2ahUKEwiQgoy5pPrrAhWKBKYKHxQCqgQ_h16BAgKEAk#imgsrc=X6EwZg-uRb0k1M

https://www.google.com/search?q=client+server+network&sxsrf=ALeKk00z4dS8Hu3k_89UUQtI129mD2EABA:1600693996487&tbm=isch&source=iu&ictx=1&fir=1on58xKWOW2dgM%252C0R8aHH6KiEU8zM%252C_&vet=1&usg=AI4_-kQq06ofi1QtuksKATOEb9WPs3TGYg&sa=X&ved=2ahUKEwiQzbfqfrAhU6K6YKHZSxBbQQ_h16BAglEAc#imgsrc=1on58xKWOW2dgM&imgdii=d7ewcDMdltrXAM

<https://techdifferences.com/difference-between-client-server-and-peer-to-peer-network.html#ComparisonChart>

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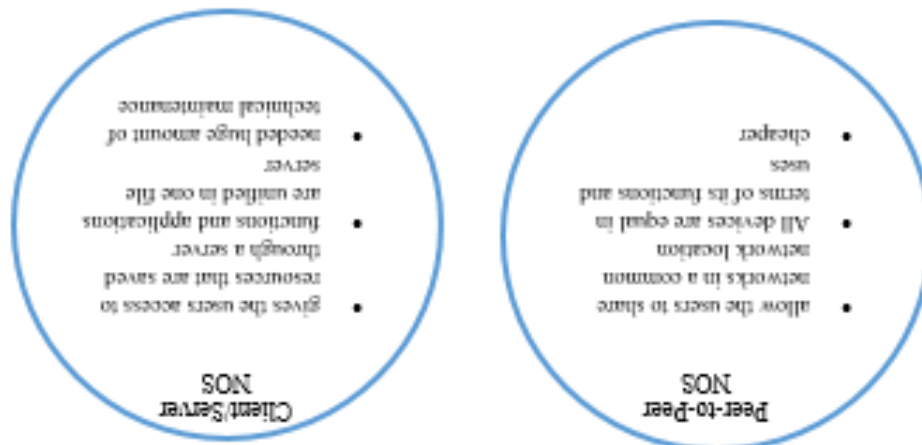
KEY TO CORRECTIONS

1	P2P	6	P2P
2	CS	7	P2P
3	CS	8	CS
4	P2P	9	P2P
5	CS	10	CS

ACTIVITY NO. 2 Comparison between Peer-to-Peer and Client Server NOS

The primary use of the network operating system is to permit shared file and printer access among multiple computers in a network, typically a local area network (LAN), whether it is a private network or to other networks.

REFLECTION



1. Network operating system is a type of operating system designed primarily to perform the function of supporting workstations, personal computers and older terminals connected on local area network also known as LAN.
2. Peer-to-peer NOS is a type of NOS that has network that is connected with each client. Each client has equal functions and uses.
3. The client/server NOS is a type of NOS that has one server connected to different client. gives the users access to resources that are saved through a server.
- 4.

ACTIVITY NO. 1 Difference/s between Peer-to-Peer and Client Server NOS